

InfDB Uruguay – summary

Status October 2026 · English · [Deutsch](#) · [Español](#) · Start: `docker compose up --build`, then <http://localhost:8050>.

Goal and status

Goal. Run the InfDB pipeline on Uruguay, starting with **Cordón**, a 228 ha barrio of Montevideo.

Done:

- inventory of Uruguayan open data and its mapping to InfDB's inputs
- a pipeline that downloads and prepares the Cordón data without modelling assumptions
- a web explorer for that data

Not done: the InfDB adaptation itself (buildings from LiDAR, attribute assignment, population distribution).

How Uruguay differs from the German InfDB inputs

InfDB input (DE)	Uruguay	Verdict
LoD2 CityGML buildings	No open building geometry. Montevideo has an open LiDAR 2024 survey (classified, building class 6). Elsewhere: Google Open Buildings 2.5D, IDE surface models 2017–18, OSM.	Main gap. Footprints and heights must be derived.
Construction year, floors, floor area estimated from Zensus grid and geometry	Observed in the cadastre. DNC publishes every <i>línea de construcción</i> monthly, nationwide: floor, use, category, condition, roof type, m², year, original year of reformed parts.	Better. InfDB SQL steps 05 and 09 become lookups.
Zensus 100 m grid	Census 2023 per census zone (13,648 polygons, Montevideo only) or segment (national). Weighted microdata with heating fuel, AC and solar water heaters per segment, after accepting INE's terms.	Polygons instead of grid. Energy variables InfDB lacks.
BKG admin units / AGS	INE and DNC codes, which differ (<code>01</code> / <code>01020</code> vs <code>V</code> / <code>AA</code>), so a crosswalk is needed.	Adapt
basemap.de streets, PLZ	IM street axes plus door points carrying the padrón and the street segment (a ready building → street link). PLZ not needed.	Equal in Montevideo
TABULA typology	None. Build from DNC category × roof × year plus census materials.	Gap
pylovo grid parameters	No UTE (utility) data	Gap
CRS 3035	EPSG:32721 (UTM 21S) everywhere. The LiDAR uses EPSG:5382 (SIRGAS-ROU98).	Config

In short: Uruguay has better building *attributes* than Germany, but no building *geometry*. InfDB's estimation steps shrink, and a LiDAR building-derivation step has to be added. Details: [comparison](#), [required changes](#).

Pilot Cordón: checked results

Data volumes:

- 5,130 parcels and 27,802 cadastral units
- 147,823 construction lines
- 287 census zones touching the barrio (186 fully inside, 35,591 inhabitants)
- 8 LiDAR tiles (164 million points, 1.2 GB)

Bulk cadastre vs. official cédula PDFs. Compared against 100 cédulas scraped from Catastro's web service:

- 99 match exactly, on every floor line, area, year, category, condition, value and the 4-year value history.
- The one difference (V-AA-137) is a 2015 reform that appears on the cédula, which is generated live, but not in the September bulk extract.
- PDF scraping is therefore unnecessary.

Joins. 99.9 % of Montevideo's cadastral parcels have a polygon. Parcel and door-point keys agree for all but 8 parcels in Cordón.

Reproducibility. A clean `docker compose up --build` downloads everything (≈ 1.5 GB) and serves the dashboard in about 9 minutes. Raw files carry SHA-256 hashes in a manifest.

Findings that need a decision before modelling

1. **Lines of two regimes on one parcel.** 917 parcels carry construction lines under both *común* (CO) and *propiedad horizontal* (PH), typically during a conversion. 40,512 lines (1.1 M m², 23 % of line area) belong to a regime without a

unit record. Summing all lines double-counts these buildings.

2. **Three "built areas" disagree.** All lines give 4.79 M m², lines of existing regimes 3.68 M m², and the units' own built area 3.07 M m². Lines include walls, landings and open areas. Which definition feeds InfDB's `floor_area`?
3. **Data quality.** 581 lines have year 0, 22,301 have 0 m², and 16,460 are *a construir* (not yet built). Some codes have no label in DNC's code table (e.g. roof code 5 on 6,252 lines).
4. **LiDAR.** The tiles are in EPSG:5382 and contain classes (11, 13, 15, 24, 105, 120, 121) that the city's documentation doesn't describe.
5. **Census zones** crossing the barrio edge hold 18,471 more inhabitants. How they are attributed is a modelling choice.

All of these are reported, flagged and filterable in the dashboard, but **not resolved**. Following the pipeline's rule, raw data is kept unchanged.

Next steps (proposed)

1. Derive building footprints and heights from the LiDAR (DTM, nDSM, footprints split by parcel).
2. Decide the rules for findings 1, 2 and 5, then adapt InfDB SQL steps 00, 02, 05–13 behind a country switch.
3. Accept INE's ANDA terms (weighted census microdata). The ingest pipeline is ready and tested on sample files.
4. Draft a Uruguayan building typology for ro-heat. Ask UTE about grid data for pylovo.
5. Extend from Cordón to barrios with detached houses (Carrasco) and informal settlements (Casavalle), then to all of Montevideo.